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# FEW-SHOT LEARNING FOR DECODING SURFACE ELECTROMYOGRAPHY FOR HAND GESTURE RECOGNITION

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## ABSTRACT

This work is motivated by the recent advancements of Deep Neural Networks (DNNs) for myoelectric prosthesis control. In this regard, hand gesture recognition via surface Electromyogram (sEMG) signals has shown a high potential for improving the performance of myoelectric control prostheses. Although the recent researches in hand gesture recognition with DNNs have achieved promising results, they are still in their infancy. The recent literature uses traditional supervised learning methods that usually have poor performance if a small amount of data is available or requires adaptation to a changing task. Therefore, in this work, we develop a novel hand gesture recognition framework based on the formulation of Few-Shot Learning (FSL) to infer the required output given only one or a few numbers of training examples. Thus in this paper, we proposed a new architecture (named as FHGR which refers to "Few-shot Hand Gesture Recognition") that learns the mapping using a small number of data and quickly adapts to a new user/gesture by combing its prior experience. The proposed approach led to $83.99\%$ classification accuracy on new repetitions with few-shot observations, $76.39\%$ accuracy on new subjects with few-shot observations, and $72.19\%$ accuracy on new gestures with few-shot observations.

***Index Terms***— Attention Mechanism, Few-Shot Learning (FSL), Meta-Learning, Temporal Convolution, sEMG.


## 1. INTRODUCTION

Recent evolution in Machine Learning (ML) and Deep Neural Networks (DNNs) coupled with advancements of neuro-rehabilitation technologies, have paved the way for the development of new control systems for myoelectric prostheses. In this regard, surface Electromyogram (sEMG) signals [1–5] is, typically, considered as the input for the control system of prostheses. In particular, the use of machine intelligence for sEMG-based Hand Gesture Recognition (HGR) has been the focus of literature due to its unique potentials to improve the functionality of control and consequently increase the quality of life of individuals with the lack of a biological limb.

Although the use of advanced machine intelligence for hand gesture recognition has been the focus of extensive academic research, the translation of these new techniques to clinical settings and to the daily lives of users is limited. Thus, there is a noticeable gap [6, 7] between practical settings and recent academic solutions in myoelectric prosthesis control, mainly due to a variety of key challenges as follows. The fist challenge is "*Variation in the sEMG Signals*" [8]. This could be caused by electrode shifts, limb position changes, and non-stationary nature of sEMG signals varying over days, muscular fatigue, skin sweating, or clinical parameters related to the amputation surgery affecting the possibility of accurate recording (e.g., level of amputation or remaining forearm percentage) [9–12]. Therefore, probability distributions of sEMG signals obtained from one user could be completely different from those of other users, and this could also be different for the same user over different days of use or condition. As a result, a trained model based on a particular set of measurements cannot directly be utilized over a long period of time. The second challenge is "***Requiring Large and Labelled Datasets***". In this regard, it can be mentioned that although DNNs perform very well when a large training dataset is available, they typically perform poorly when the available data is limited. Large-scale data collection maybe possible in a laboratory setting but is not feasible in real life. Addressing the above challenges can significantly increase the practicality of the myoelectric prosthesis.

Therefore, there is a need to employ adaptive DNN-based learning frameworks, which focus on transferring knowledge from a source domain to a target domain, despite the presence of a non-stationary mismatch between them (i.e., to tackle the first challenge). This will also allow achieving high performance without requiring large amounts of data (i.e., to tackle the second challenge).

**1.1. Literature Review**: A common procedure to identify hand movements with DNN-based algorithms is to use $2/3$ of the gesture trials from each subject for training purposes. The remaining $1/3$ of trials constituting the test set are then utilized to evaluate the model [13–17]. Although recent academic researchers are yielding high performances, the training process requires a significant amount of new training data. If there is a discrepancy between the training and the test phase, such methods may not work properly, and a large amount of data collection should be conducted for a new gesture or a user. Therefore, to bridge the gap between usage in real-world and scientific developments under laboratory conditions, some literature take steps towards domain adaptation approaches. Reference [10], for instance, compensates the effects of electrode shifts and avoids large data requirements by applying Transfer Learning (TL) from the image recognition domain to the High Density-sEMG (HD-sEMG) classification domain. In addition, TL-based algorithms have also been used to leverage the information and knowledge gained from multiple users and speed up the learning process for new users [18, 19]. In addition, some research (.e.g [20, 21]) use domain adaption techniques to introduce robustness against the day-to-day variant nature of the sEMG signals. More specifically, instead of recording sEMG signals in short sessions, they collect sEMG signals in multiple days (e.g., two sessions per day over 15 days) and then used these signals to develop and test DNNs networks.

All these research papers have evaluated the performance of

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their proposed methods under a strongly controlled environment with limited adaptability power. Therefore, translation of these conventional models to the clinical settings with a large number of users and gestures is debatable. Thus, more research is required on the development of adaptive learning techniques to train the model only based on few new examples. Indeed, this is a challenging task given neurophysiology differences between users [4, 7] and consequences of amputations. The paper aims to take a step towards tackling this challenge.

**1.2. Contributions**: In this paper, we propose a novel and innovative Few-Shot Learning (FSL) framework referred to as the FHGR for hand-gesture recognition. The proposed FHGR relies on minimal new data and allows the user to retrain the DNN framework by looking at a few examples from each class. More specifically, the proposed FHGR architecture takes advantage of domain adaption, which decreases the need for recalibration and decoding of new gestures just by a small amount of data (when compared to its traditional counterparts). The contributions of the paper are summarized below: (i) The FHGR is designed by combining temporal convolutions and attention mechanisms, to reduce the required training time and to mitigate the challenge of variability; (ii) The FHGR eliminates the need for extensive recalibration via FSL techniques in which knowledge will be acquired quickly from a few basic exercises; (iii) The FHGR adapts, relying on a small number of new observations of new sEMG signals, and (iv) The FHGR accelerates the learning process for a new user/gesture by utilizing knowledge gained from previous source users/gestures. The learning process does not start every time from scratch resulting in a faster adaptation pace.

## 2. THE PROPOSED FHGR ARCHITECTURE

Developing FSL algorithms for hand gesture recognition can tackle the challenges in Section 1 to improve myoelectric prosthesis control. Most of the FSL approaches, however, are proposed for image classification tasks [22–24]. Inspired by [24], we extend the FSL to sEMG time-series classification and propose the FHGR framework. In what follows, first, the pre-processing step is discussed followed by definition of meta-learning, also known as *learning to learn*. Then, we develop the proposed few-shot classification FHGR model for sEMG signals.

### 2.1. Preprocessing Step

The raw time-domain sEMG signals and time-channel-frequency spectrograms are the two inputs for the FHGR architecture. For utilizing raw sEMG as the input, we follow the pre-processes approach in the existing literature [9, 16, 25, 26] and use a 1st order low-pass Butterworth filter to smooth the sEMG signals. Moreover, for scaling the sEMG signals, the $\mu$-*law* transformation is applied to sensors with small values, amplifying their output in a logarithmic fashion [1]. This transformation keeps the scale of sensors with larger magnitudes over time. The following formulation describes the $\mu$-*law* transformation, used traditionally in speech area for quantization but is used here for sEMG signal processing

$$F(x_t) = \text{sign}(x_t)\frac{\ln\big(1+\mu|x_t|\big)}{\ln\big(1+\mu\big)}, \tag{1}$$

where $t \geq 1$ denotes the time; $x_t$ shows the sEMG signal to be scaled, and $\mu$ is the parameter defining new range. Next, the Minmax normalization is applied to the scaled inputs. It has been experimentally observed that scaling sEMG signals with the mentioned preprocessing approach leads to better output. The spectrogram is calculated using a 256-point Fast Fourier Transform (FFT) with a Hamming window and hop-length of 72. This procedure converts

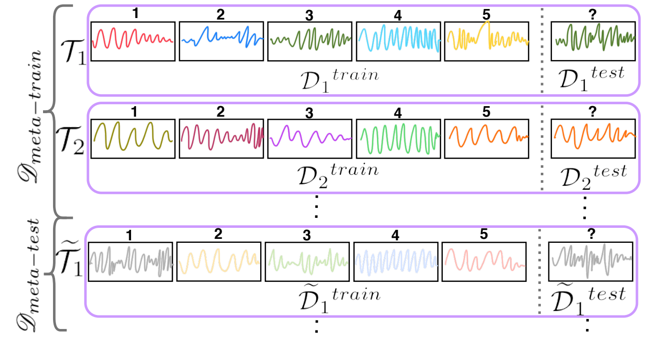


**Fig. 1**: Example of meta-leaning setup for HGR. $\mathscr{D}_{meta-train}$ is used for training purposes, where each task $\mathcal{T}_j$ is drawn from $p\left(\mathcal{T}\right)$. Set $\mathscr{D}_{meta-test}$ consists of unseen tasks sampled from a different distribution (i.e., $\widetilde{\mathcal{T}}_j \sim p(\widetilde{\mathcal{T}})$), where $p(\widetilde{\mathcal{T}})$ is similar in nature to $p\left(\mathcal{T}\right)$. Moreover, $\mathscr{D}_{meta-val}$, defined similarly, is used to determine the hyper-parameters of the model. Task $\mathcal{T}_j$, represented in the purple box, is associated with a dataset $\mathcal{D}$ splitting into two parts: training set $\mathcal{D}^{train}$ and prediction set $\mathcal{D}^{test}$. Here, we are illustrating 5-way 1-shot classification where $\mathcal{D}^{train}$ contains $k = 1$ labelled example for each of $N = 5$ classes (each class represented with a different colour and is associated with a label 1-5). $\mathcal{D}^{test}$ consists of 1 unlabelled example, which is sampled from one of those $N = 5$ classes to be predicted by the network.

each segment of raw sEMG to a spectrogram with 129 frequencies covering the range ($0 - 1,000$ Hz). Then and following the procedure in [18, 27], to decrease the baseline drift, the first frequency is removed. Moreover, we removed the frequencies in range (700-$-\ 1,000$ Hz) because the majority of the sEMG energy does not lie in this band. This completes the pre-processing step to prepare raw sEMG and spectrogram inputs. Next, we describe the proposed FSL-based FHGR model.

### 2.2. Meta-Learning Structure of the FHGR

In meta-learning, unlike supervised-learning approaches that generalize across data points, the goal is generalizing the model to the new tasks having seen only a few numbers of updates. Therefore, instead of using the traditional formulation for supervised-learning methods, we are dealing with task $\mathcal{T}_j \in \mathscr{D}$ which is episodic and associated with a dataset $\mathcal{D}_j$. As shown in Fig. 1, task $\mathcal{T}_j$ has two components: a train-set $\mathcal{D}_j{}^{train}$ for learning, and a test-set $\mathcal{D}_j{}^{test}$ for evaluation, i.e., $\mathcal{T}_j = (\mathcal{D}_j{}^{train}, \mathcal{D}_j{}^{test})$. In such a meta-learning formulation, $\mathscr{D}$ indicates meta-set, which could be meta-train $\mathscr{D}_{meta-train}$, or meta-test $\mathscr{D}_{meta-test}$. In $\mathscr{D}_{meta-train}$, a meta-learner is trained in such a way that in each episode, it receives one of its training sets $\mathcal{D}_j{}^{train} \in \mathcal{T}_j$ and produces a learner that could predict the expected output on its corresponding test-set $\mathcal{D}_j{}^{test}$. $\mathscr{D}_{meta-test}$ is used for evaluating the model generalization capacity. It is noteworthy to say that the tasks in the meta-sets cover disjoint sets of classes, i.e., the classes presented in any $\mathcal{T}_j$ in $\mathscr{D}_{meta-train}$ are not included in any task in $\mathscr{D}_{meta-test}$ and vice versa.

Within the context of meta-learning, we are interested to focus on FSL formulation where the goal is reducing the prediction error on unlabelled examples given a small training set. Following the notation of [24], we define $N$-way $k$-shot problem such that for each task $\mathcal{T}_j$, the training set consists of a sequence of example-label pairs $\mathcal{D}_j{}^{train} = \{(\mathbf{x}_i, \mathrm{y}_i)\}_{i=1}^{k\times N}$, followed by $\mathcal{D}_j{}^{test}$, which consists of an unlabelled example. The FSL model aims to predict the label of the final example based on the previous labels that it has seen. To be more specific, in our $N$-way $k$-shot classification task, $\mathcal{D}_j{}^{train}$ consists of $k$ examples for each of $N$ unique classes and $\mathcal{D}_j{}^{test}$ has an unlabelled example sampled from $N$ selected classes. Fig. 1 illustrates 5-way 1-shot classification task. Each example is concate-

**Algorithm 1** THE TRAINING PROCEDURE

**Input:** $\mathscr{D}_{meta-train}$, and; mapping function $f(\cdot)$ with parameters $\theta$.
Require. $p(\mathcal{T})$: distribution over tasks
**while** not done **do**
  Sample batch of tasks $\mathcal{T}_j \sim p(\mathcal{T})$
  **for all** $\mathcal{T}_j$ **do**
    Split $\mathcal{T}_j$ into $\mathcal{D}_j{}^{train}$ and $\mathcal{D}_j{}^{test}$
    Predict the missing label of final example of $\mathcal{T}_j$: $\hat{\mathrm{y}}_{\mathrm{test}} = f(\mathcal{D}_j{}^{train}, \mathcal{D}_j{}^{test}; \theta)$
  **end for**
  Update $\theta$ using $\Sigma_{\mathcal{T}_j \sim p(\mathcal{T})} \mathcal{L}_{\mathcal{T}_j}(\hat{y}_{test}, y_{test})$
**end while**

nated with its corresponding label (the unlabelled example in $\mathcal{D}_j{}^{test}$ is concatenated with a null label). Then, these $(N \times k + 1)$ example-label pairs are sequentially fed as the input to network. Finally, for training purposes, we compute the loss $\mathcal{L}_j$ between the ground truth and predicted label of the $(N \times k + 1)^{\text{th}}$ example (Algorithm 1).

### 2.3. The FHGR Architecture

The proposed FHGR framework adopts Temporal Convolutions (TC) coupled with attention mechanisms. More specifically, by using Dilated Causal 1D-Convolutions, the model provides several benefits over Recurrent Neural Networks (RNNs) such as easier training, simpler and clearer architecture. In addition, by using larger dilation at the top level instead of increasing the number of layers, the output can access a broad range of past experiences [24, 28, 29]. Moreover, use of attention mechanisms allows the model to identify particular information within a large context [30]. As illustrated in Fig. 2, the FHGR framework has been developed based on four modules, namely *Embedding Module*, *TBlock Module*, *TCNet Module*, and *Attention Module*, which are described below:

(i) *The Embedding Module*: To make multidimensional input compatible with the proposed FHGR architecture, we develop the Embedding Module, which repeats the following block several times $\{3 \times 3$ conv, batch norm, ReLU, Max Pool 2D$\}$. As shown in Fig. 2, this module (the gray box) extracts a 128-dimensional feature vector from raw sEMG signals or corresponding spectrogram. Each feature vector is then concatenated with its corresponding label to form the input for the next stage. It is noteworthy to mention that raw sEMG signals acquired from $N_S$ number of sensors which are segmented by a window of length of $W = 200$ ms. We adopt three Embedding Modules, namely *Spect-Embed1*, *Raw-Embed*, and *Spect-Embed1*2 as detailed in Table 1.

(ii) *TBlock Module*: Each TBlock Module, represented via a green box in Fig. 2, consists of two Dilated Causal 1D-Convolutions (DC-conv), each with dilation factor $d$, kernel size $k$, and $f$ number of channels. After each DCconv, a ReLU activation function is included to better learn the complex structure of the underlying data. Finally, results and the input are concatenated leading to improved overall performance. More specifically, the TBlock Module takes an input of size (input-dimensionality $\times$ $T$) and outputs a tensor of size (output-dimensionality $\times$ $T$). Here, $T = N \times k + 1$ denotes the sequence length, where $N$ is the number of class samples from the set of labels, and $k$ denotes the number of examples per each class.

(iii) *TCNet Module*: The TCNet Module, represented via a yellow box in Fig. 2, consists of a series of TBlock Module in which dilation factor $d$ grows exponentially, i.e., $(1, 2, 4, 8, \ldots)$, to access extended history. More specifically, the TCNet Module consists of $Z = \lceil \log_2 T \rceil$ number of TBlock Module for an input with sequence length of $T = N \times k + 1$ in the $N$-way $k$-shot classification problem.

(iii) *The Attention Module*: An attention block performs key-query similarity measurement (e.g., dot product) to compute weighted average of the values [30]. The keys, values, and queries are packed together into matrices $\boldsymbol{K}$, $\boldsymbol{V}$, and $\boldsymbol{Q}$, respectively, resulting in

$$\text{Attention}(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}) = \text{softmax}(\frac{\boldsymbol{Q}\boldsymbol{K}^T}{\sqrt{d_k}})\boldsymbol{V}, \quad (2)$$

where $d_k$ stands for length of the key vector in matrix $\boldsymbol{K}$. Finally, residual connections are used to concatenate the output and input.

**Table 1**: Parameters used in design of *Spect-Embed1*, *Raw-Embed*, and *Spect-Embed1*2. Each module repeats the following block for 3 or 4 times $\{3 \times 3$ conv, batch norm, ReLU, MaxPool 2D$\}$.

| Name | Input | # Blocks | # Channels in each block | Kernel-size of MaxPool in each block |
|---|---|---|---|---|
| Spect-Embed1 | Spectrogram | 4 | 128, 128, 128, 128 | (1,3), (2,3), (2,3), (2,3) |
| Raw-Embed | Raw sEMG | 3 | 8, 16, 32 | (1,2), (3,4), (2,25) |
| Spect-Embed2 | Spectrogram | 3 | 8, 16, 32 | (1,2), (3,4), (2,5) |

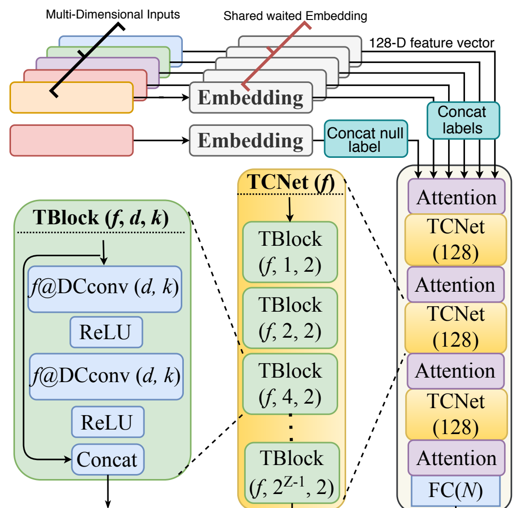


**Fig. 2**: The proposed FHGR architecture for $N$-way $k$-shot classification consisting of four Attention Modules, three TCNet Module, and a Fully Connected (FC) layer with output size of $N$. Each TCNet Module consist of $Z = \lceil \log_2 T \rceil$ number of TBlock Modules. In TBlock Module, *f*@DCconv(*d*, *k*) stands for *f* number of Dilated Causal 1D-Convolution kernels with dilation factor $d$ and kernel size $k = 2$. More specifically, for each task $\mathcal{T}_j$, first, each multi-dimensional input is passed through Embedding Module, where a 128-dimensional feature vector is extracted. Then, labels corresponding to the inputs are concatenated to each feature vector. The final input in the sequence is concatenated with a null label to be predicted by the network.

## 3. EXPERIMENTS AND RESULTS

In this section, first, we describe the database used to evaluate the proposed FHGR, followed by different experiments and results.

### 3.1. Database

In this work, the second Ninapro dataset [9] referred to as the DB2 is used. DB2 dataset is built by Delsys Trigno Wireless EMG system with 12 wireless electrodes (channels), recording muscular activity at a rate of 2 kHz. More specifically, the DB2 dataset consists of signals collected from 40 intact-limb users (28 males and 12 females with age 29.9±3.9 years among which 34 are right-handed and 6 are left-handed) performing 50 gestures including wrist, hand, grasping, and functional movements. Each movement in the DB2 dataset is repeated 6 times, each lasting for 5 sec. followed by 3 sec. of rest. The DB2 dataset was presented in three sets of exercises denoted by Exercise $B$, $C$, and $D$. Please refer to [9] for more details.

**Table 2**: *Experiment 1*: 5-way, 1-shot and 5-shot classification accuracies on new repetitions with few-shot observations, performed by using meta-learning approach. The table also shows a comparison between the proposed FHGR Meta-supervised learning methodology and previous works where Supervised learning is used.

| | | The Embedding Module | 5-way Accuracy 1-shot | 5-way Accuracy 5-shot |
|---|---|---|---|---|
| Meta Learning | Proposed FHGR | Spect-Embed1 | 70.71% | 80.40% |
| | | Raw-Embed | **71.23**% | **83.99**% |
| | | Spect-Embed2 | 61.51% | 74.54% |
| Supervised Learning | Previous Works | **Previous Works** | **Accuracy** | |
| | | Wei *et al.* [13] | **83.70**% | |
| | | Hu et al. [14] | 82.20% | |
| | | Ding *et al.* [31] | 78.86% | |
| | | Zhai *et al.* [27] | 78.71% | |
| | | Geng *et al.* [25] | 77.80% | |
| | | Atzori *et al.* [26] | 75.27% | |

**Table 3**: Experiment 2: 5-way, 1-shot and 5-shot classification accuracies based on *new subjects with few-shot observation*.

| The Embedding Module | 5-way Accuracy | |
|---|---|---|
| | 1-shot | 5-shot |
| Spect-Embed1 | 52.31% | 69.61% |
| Raw-Embed | **66.45**% | **76.39**% |
| Spect-Embed2 | 54.64% | 70.77% |

**Table 4**: Experiment 3: 5-way, 1-shot and 5-shot classification accuracies based on *new gesture with few-shot observation*.

| The Embedding Module | 5-way Accuracy | |
|---|---|---|
| | 1-shot | 5-shot |
| Spect-Embed1 | 38.35% | 57.1% |
| Raw-Embed | **47.28**% | **72.19**% |
| Spect-Embed2 | 38.78% | 58.06% |

### 3.2. Results and Discussions

In the following reported experiments, Adam optimizer was used for training purposes with learning rate of $0.0001$. Different models were trained with a mini-batch size of 64. The average loss is computed using Cross-entropy loss and reported on the $(N \times k + 1)^{th}$ example. Three evaluation experiments are considered as follows:

***Experiment 1 - Classification on New-Repetitions with Few-Shot Observations***: This experiment is included to evaluate performance of the proposed FHGR when faced with new repetitions with few-shot observations on the target. Existing state-of-the-art research works [13, 14, 25–27, 31] use approximately $2/3$ of the gesture trials of each subject for training purposes. Following these works, therefore, in this experiment we evaluate the FHGR when $\mathscr{D}_{meta-train}$ consists of the $2/3$ of the gesture trials of each subject (following Reference [26], repetitions 1, 3, 4, and 6 are used for training), and $\mathscr{D}_{meta-test}$ consists of the remaining repetitions. Table 2 shows the results associated with the proposed few-shot classification approach and previous supervised learning-based methodologies. From Table 2, it can be observed that the proposed FHGR architecture outperforms existing methodologies when evaluated based on the same setting, i.e., 83.99% best accuracy from the FHGR compared to 83.70% best accuracy achieved by the state-of-the-art.

***Experiments 2 - Classification on New-Subject with Few-Shot Observations***: In this experiment, to show that the proposed FHGR architecture can classify hand gestures of new subjects just by training with a few examples, we split the DB2 database into $\mathscr{D}_{meta-train}$, $\mathscr{D}_{meta-val}$, and $\mathscr{D}_{meta-test}$ such that the subjects in these meta-sets are completely different (i.e., there is no overlap between the meta-sets). More specifically, $\mathscr{D}_{meta-train}$ consists of the first 27 subjects, while $\mathscr{D}_{meta-val}$ includes the sEMG signals from the $28^{th}$ subject to $33^{rd}$ subject. Finally, we evaluated our model on the remaining 8 subjects. It is noteworthy to mention that the proposed network is trained once and shared across all participants (this is in contrary to some existing works that train the model separately for each participant). For constructing each task $\mathcal{T}_j$, we constrained ourselves to sample all of $N$ classes from a specific user, randomly selected from the existing participants, which is a more realistic and practical scenario. Table 3 shows the results for Experiment 2 based on three Embedding Modules. The results of Table 3 show that the proposed FHGR achieved acceptable results in transferring information between a source and a target domain despite the existence of a distribution mismatch between them.

***Experiment 3 - Classification on New-Gestures with few-shot Observations***: Here, the objective is to evaluate capability of the proposed FHGR architecture when the target consists of solely out-of-sample gestures (i.e., a new gesture with few shot observations). Performing well in this task allows the model to evaluate new observations, exactly one per novel hand gesture class. For training purposes, $\mathscr{D}_{meta-train}$ consists of the first 34 gestures of each user, which is equal to approximately $68\%$ of the total gestures. $\mathscr{D}_{meta-val}$ includes 6 gestures or $12\%$ of the total gestures. The remaining gestures (9 gestures), were used in $\mathscr{D}_{meta-test}$ for evaluation purposes. Exercises $B$ and $C$ were, therefore, used for training and validation, and Exercises $D$ with totally different gestures were used for test purposes. Table 4 shows the efficiency of the FHGR when provided with out-of-sample gestures. It can be observed that the model can predict unknown class distributions in scenarios where few examples from the target distribution is available.

***Effects of Different Embedding Modules***: In the three experiments, the network has better performance when it receives the raw sEMG signal as the input. The Raw-Embed has similar structure to that of the Spect-Embed2, and the main difference is the input format. Therefore, it can be concluded that the network itself can extract more informative features, improving the overall performance. Moreover, in Experiment 1, where $\mathscr{D}_{meta-train}$ and $\mathscr{D}_{meta-test}$ have the same distribution, the Spect-Embed1 has better performance than Spect-Embed2. The intuition is that deeper structure of Spect-Embed1 helps it to mimic data distribution better than Spect-Embed2. In contrast, as shown in Experiments 2 and 3, the performance of Spect-Embed1 drops in comparison with Spect-Embed2. The main reason is that data distribution in $\mathscr{D}_{meta-train}$ and $\mathscr{D}_{meta-test}$ is not the same anymore. Therefore, the simpler structure in Spect-Embed2 contributes to its generalization resulting in better performance on the test-set having a different distribution from that of the train-set.

## 4. CONCLUSION

In this paper, we proposed a novel few-shot learning recognition approach, the FHGR framework, for HGR tasks via sEMG signals. The FHGR architecture can quickly generalize after seeing few examples from each class, which is achieved by exploiting the knowledge gathered from previous experiences. The ability to learn quickly based on a few examples is a key feature of the proposed FHGR framework that distinguishes this novel architecture from its previous counterparts. Additionally, the proposed FHGR framework transfers information between a source and a target domain despite the existence of a distribution mismatch among them. Thus, the learning process does not start every time from scratch, and instead refines. This would reduce the number of required training sessions, which in turn leads to a potential reduction in functional prosthesis abandonment.

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